

Chinese Remainder Theorem Exercises

Q&A Notes

Question 1

Find all solutions to the equations

$$x \equiv 4 \pmod{5}$$

and

$$x \equiv 5 \pmod{11}.$$

Answer 1

The solution is

$$x \equiv 49 \pmod{55}.$$

So all integer solutions are

$$x = 49 + 55t, \quad t \in \mathbb{Z}.$$

The moduli 5 and 11 are relatively prime, so the Chinese Remainder Theorem applies. First compute the combined modulus:

$$N = 5 \cdot 11 = 55.$$

For the congruence modulo 5:

$$M_1 = \frac{55}{5} = 11.$$

We need y_1 such that

$$11y_1 \equiv 1 \pmod{5}.$$

Since

$$11 \equiv 1 \pmod{5},$$

we can take

$$y_1 = 1.$$

For the congruence modulo 11:

$$M_2 = \frac{55}{11} = 5.$$

We need y_2 such that

$$5y_2 \equiv 1 \pmod{11}.$$

Since

$$5 \cdot 9 = 45 \equiv 1 \pmod{11},$$

we can take

$$y_2 = 9.$$

Now combine:

$$x \equiv 4M_1y_1 + 5M_2y_2 \pmod{55}.$$

$$x \equiv 4(11)(1) + 5(5)(9) \pmod{55}.$$

$$x \equiv 44 + 225 \pmod{55}.$$

$$x \equiv 269 \pmod{55}.$$

$$269 = 55(4) + 49.$$

Therefore

$$x \equiv 49 \pmod{55}.$$

Check:

$$49 \equiv 4 \pmod{5}, \quad 49 \equiv 5 \pmod{11}.$$

Question 2

Find all integers x that leave remainders 1, 2, 3 when divided by 9, 8, 7, respectively.

Answer 2

The solution is

$$x \equiv 10 \pmod{504}.$$

So all integer solutions are

$$x = 10 + 504t, \quad t \in \mathbb{Z}.$$

The system is

$$x \equiv 1 \pmod{9}, \quad x \equiv 2 \pmod{8}, \quad x \equiv 3 \pmod{7}.$$

The moduli 9, 8, 7 are pairwise relatively prime, so CRT applies.

Compute

$$N = 9 \cdot 8 \cdot 7 = 504.$$

Congruence	n_i	a_i	$M_i = N/n_i$
1	9	1	56
2	8	2	63
3	7	3	72

Now find the inverses.

For $M_1 = 56$, work modulo 9:

$$56 \equiv 2 \pmod{9}.$$

We need

$$2y_1 \equiv 1 \pmod{9}.$$

Since

$$2 \cdot 5 = 10 \equiv 1 \pmod{9},$$

we take

$$y_1 = 5.$$

For $M_2 = 63$, work modulo 8:

$$63 \equiv 7 \pmod{8}.$$

We need

$$7y_2 \equiv 1 \pmod{8}.$$

Since

$$7 \cdot 7 = 49 \equiv 1 \pmod{8},$$

we take

$$y_2 = 7.$$

For $M_3 = 72$, work modulo 7:

$$72 \equiv 2 \pmod{7}.$$

We need

$$2y_3 \equiv 1 \pmod{7}.$$

Since

$$2 \cdot 4 = 8 \equiv 1 \pmod{7},$$

we take

$$y_3 = 4.$$

Combine:

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \pmod{504}.$$

$$x \equiv 1(56)(5) + 2(63)(7) + 3(72)(4) \pmod{504}.$$

$$x \equiv 280 + 882 + 864 \pmod{504}.$$

$$x \equiv 2026 \pmod{504}.$$

Reduce:

$$2026 = 504(4) + 10.$$

Therefore

$$\boxed{x \equiv 10 \pmod{504}.$$

Check:

$$10 \equiv 1 \pmod{9}, \quad 10 \equiv 2 \pmod{8}, \quad 10 \equiv 3 \pmod{7}.$$

Question 3

Argue that, under the definitions of the Chinese Remainder Theorem, if $\gcd(a, n) = 1$, then the inverse of a modulo n corresponds to the inverses of its parts modulo $n_1, n_2, \dots, n_k$.

Answer 3

The inverse modulo n reduces to the inverse in each smaller modulus.

Assume

$$n = n_1 n_2 \cdots n_k,$$

where the n_i 's are pairwise relatively prime.

Let

$$b \equiv a^{-1} \pmod{n}.$$

This means

$$ab \equiv 1 \pmod{n}.$$

If a number is congruent to 1 modulo n , then it is also congruent to 1 modulo every divisor n_i of n . Hence

$$ab \equiv 1 \pmod{n_i}$$

for every i .

Now reduce a and b modulo n_i . Write

$$a_i \equiv a \pmod{n_i}, \quad b_i \equiv b \pmod{n_i}.$$

Then

$$a_i b_i \equiv 1 \pmod{n_i}.$$

So b_i is the inverse of a_i modulo n_i .

Therefore

$$\boxed{a^{-1} \pmod{n} \text{ corresponds to } (a_1^{-1} \pmod{n_1}, \dots, a_k^{-1} \pmod{n_k}).}$$

Why this makes sense

CRT lets us work modulo n by working separately modulo $n_1, n_2, \dots, n_k$. Multiplicative inverses also split this way.

Question 4

Under the definitions of the Chinese Remainder Theorem, prove that for any polynomial f , the number of roots of

$$f(x) \equiv 0 \pmod{n}$$

equals the product of the numbers of roots of

$$f(x) \equiv 0 \pmod{n_1}, \quad f(x) \equiv 0 \pmod{n_2}, \quad \dots, \quad f(x) \equiv 0 \pmod{n_k},$$

where $n = n_1 n_2 \cdots n_k$ and the n_i 's are pairwise relatively prime.

Answer 4

The number of roots modulo n is the product of the numbers of roots modulo the smaller moduli.

Let

$$R_i = \{r \pmod{n_i} : f(r) \equiv 0 \pmod{n_i}\}$$

be the set of roots modulo n_i .

Suppose

$$x \equiv r_i \pmod{n_i}$$

for each i . By CRT, every choice of residues

$$(r_1, r_2, \dots, r_k)$$

comes from exactly one residue x modulo n .

Now,

$$f(x) \equiv 0 \pmod{n}$$

is equivalent to saying

$$n \mid f(x).$$

Since

$$n = n_1 n_2 \cdots n_k$$

and the n_i 's are pairwise relatively prime, this is equivalent to

$$n_i \mid f(x)$$

for every i . In congruence form:

$$f(x) \equiv 0 \pmod{n_i} \quad \text{for every } i.$$

So x is a root modulo n exactly when its residue modulo each n_i is a root modulo n_i .

Therefore the roots modulo n are obtained by independently choosing:

$$r_1 \in R_1, \quad r_2 \in R_2, \quad \dots, \quad r_k \in R_k.$$

If

$$|R_1| = s_1, \quad |R_2| = s_2, \quad \dots, \quad |R_k| = s_k,$$

then the number of possible choices is

$$s_1 s_2 \cdots s_k.$$

Hence

$$\#\{x \pmod{n} : f(x) \equiv 0 \pmod{n}\} = \prod_{i=1}^k \#\{x \pmod{n_i} : f(x) \equiv 0 \pmod{n_i}\}.$$